

The page features a decorative border with a black and gray checkered pattern on the left and right sides. The top corners are rounded. At the bottom, there are stylized cloud shapes in light gray and dark gray. The main title is centered on a white rectangular background.

2nd Quarter: Activities

Reverse String

```
1 package reversestring;
2 import java.util.Scanner;
3
4 public class ReverseString {
5     public static void main(String[] args) throws InterruptedException {
6         String input;
7         StringBuilder r = new StringBuilder();
8         Scanner me = new Scanner(System.in);
9
10        System.out.print("Enter a name > ");
11        input = me.next();
12
13        for (int i = input.length() - 1; i >= 0; i--) {
14            r.append(input.charAt(i));
15        }
16        System.out.println(r);
17        for (int i = 0; i < input.length(); i++) {
18            System.out.println(input.charAt(i));
19            Thread.sleep(500);
20        }
21    }
22 }
```

Output - ReverseString (run) X

```
run:
Enter a name > PROGRAMMING
GNIMMARGORP
P
R
O
G
R
A
M
M
I
N
G
BUILD SUCCESSFUL (total time: 22 seconds)
```

Pyramid

```
1 package pyramid;
2
3 public class Pyramid {
4
5     public static void main(String[] args) {
6         int height = 7;
7         StringBuilder r = new StringBuilder();
8
9         for(int i = 1; i<=height; i++){
10             for(int a = 1; a<=height-i; a++){
11                 r.append(" ");
12             }
13             for(int b = 1; b<=(2*i-1); b++){
14                 r.append("#");
15             }
16             r.append("\n");
17         }
18         System.out.println(r.toString());
19     }
20 }
21
```

Output - Pyramid (run) X

run:

```

      #
     ###
    #####
   #####
  #####
 #####
#####
#####

BUILD SUCCESSFUL (total time: 0 seconds)
```

Business Letter

```
1 package simulation;
2 import javax.swing.*;
3
4 public class Simulation {
5     public static void main(String[] args) {
6         String name = JOptionPane.showInputDialog(null, "Please enter your first and last name.");
7         String fn = JOptionPane.showInputDialog(null, "Enter the name of the receiver of the letter.");
8         String address = JOptionPane.showInputDialog(null, "Enter your address.");
9         String gmail = JOptionPane.showInputDialog(null, "Enter your gmail.");
10        String contact = JOptionPane.showInputDialog(null, "Enter your contact number.");
11        String date = JOptionPane.showInputDialog(null, "Enter the date today.");
12        int x = 0;
13        char c;
14
15        while (x < fn.length()) {
16            if (fn.charAt(x) == ' ') {
17                fn = fn.substring(0, x);
18                x = fn.length();
19            }
20            ++x;
21        }
22
23        JOptionPane.showMessageDialog(null, name + "\n" + address + "\n" + gmail + "/" + contact + "\n" + date +
24            "\n\nDear " + fn + ",\nI deeply apologize for inconvenience that I have caused "
25            + "yesterday.\nI was not present for the reason that I had an emergency. "
26            + "\nI hope this letter will find you and you will "
27            + "forgive me. \nThank you." + "\n\nSincerely,\n" + name);
28    }
29 }
```

Input

?

Please enter your first and last name.

Melody Payod

OK Cancel

Input

?

Enter the name of the receiver of the letter.

Sir B

OK Cancel

Input

?

Enter your address.

12 Somewhere

OK Cancel

Input

?

Enter your gmail.

name@gmail.com

OK Cancel

Input

?

Enter your contact number.

09123456789

OK Cancel

Input

?

Enter the date today.

11/23/2007

OK Cancel

Message

i

Melody Payod
12 Somewhere
name@gmail.com/09123456789
11/23/2007

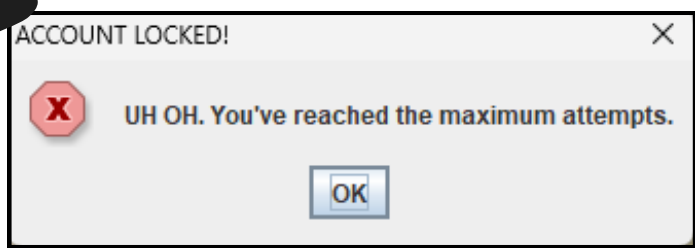
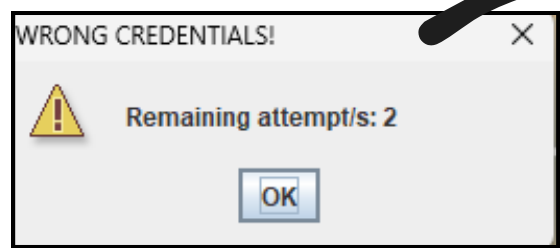
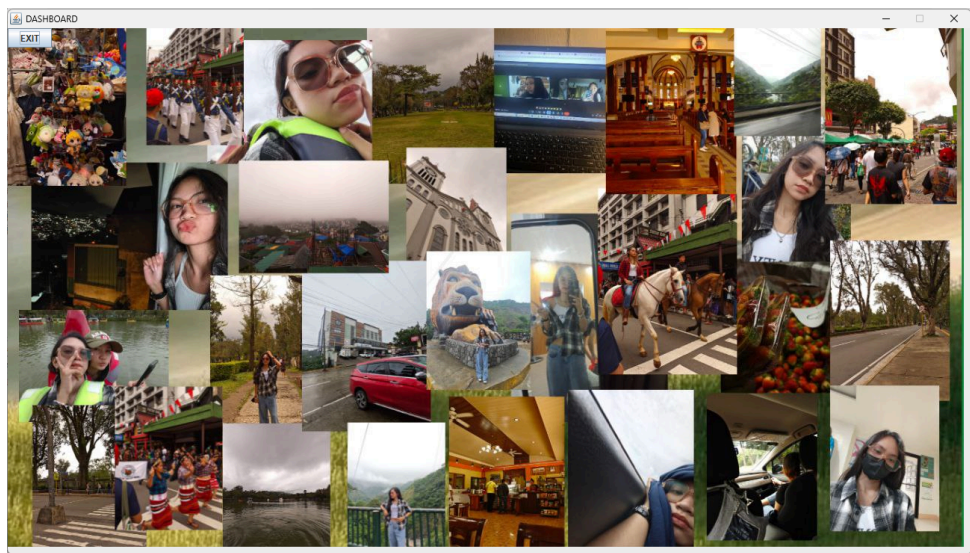
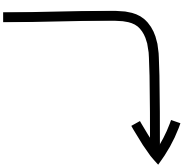
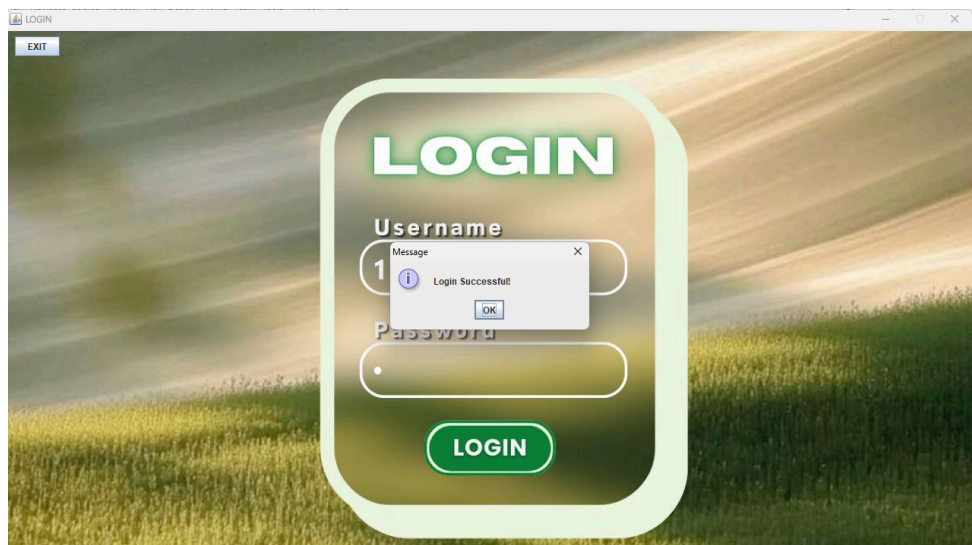
Dear Sir,
I deeply apologize for inconvenience that I have caused yesterday.
I was not present for the reason that I had an emergency.
I hope this letter will find you and you will forgive me.
Thank you.

Sincerely,
Melody Payod

OK



Login GUI (StringBuilder)



Try again in 26 seconds.

Login Console (StringBuilder)

```
Source History
2 import java.util.Scanner;
3 public class LoginStringBuilder {
4     public static void main(String[] args) {
5         Scanner meld = new Scanner(System.in);
6         StringBuilder user = new StringBuilder("Melody");
7         StringBuilder pass = new StringBuilder("2311");
8         int input;
9         String NewUser, NewPass, cmpUser, cmpPass;
10
11         System.out.println("Simple Login System Using StringBuilder Only");
12
13         do{
14             System.out.println("\nMENU\n1.Register\n2.Login\n3.Exit\nChoose an option: ");
15             input=meld.nextInt();
16             if(input == 1){
17                 System.out.print("\nREGISTER\nEnter new username: ");
18                 NewUser=meld.next();
19                 user.setLength(0);
20                 user.append(NewUser);
21                 System.out.print("Enter new password: ");
22                 NewPass=meld.next();
23                 pass.setLength(0);
24                 pass.append(NewPass);
25                 System.out.print("***REGISTRATION COMPLETE!\n=====\\n");
26             }
27             if(input==2){
28                 System.out.print("\nLOGIN\nEnter username: ");
29                 cmpUser=meld.next();
30                 System.out.print("Enter password: ");
31                 cmpPass=meld.next();
32                 if(cmpUser.equals(user.toString()) && cmpPass.equals(pass.toString()))
33                     System.out.println("***LOGIN SUCCESS!\n=====\\n");
34                 else
35                     System.out.println("LOGIN FAILED!\\n");
36             }
37         } while (input!=3);
38     }
```

```
run:
Simple Login System Using StringBuilder Only

MENU
1.Register
2.Login
3.Exit
Choose an option:
1

REGISTER
Enter new username: ody
Enter new password: 1234
***REGISTRATION COMPLETE!
=====

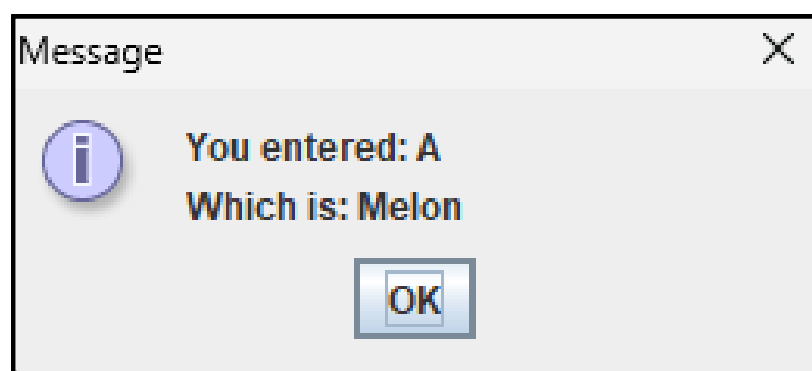
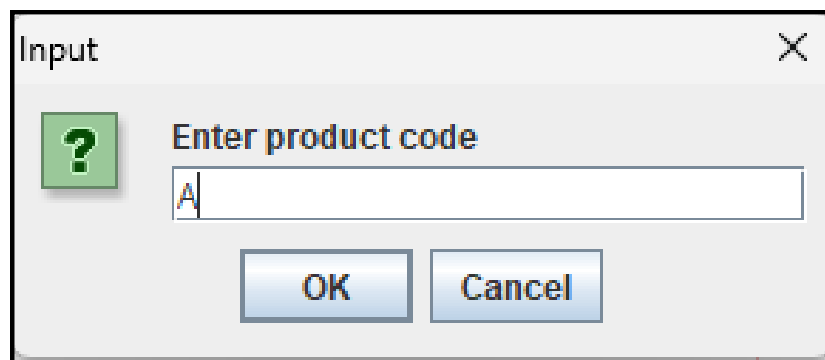
MENU
1.Register
2.Login
3.Exit
Choose an option:
2

LOGIN
Enter username: ody
Enter password: 1234
***LOGIN SUCCESS!
=====

MENU
1.Register
2.Login
3.Exit
Choose an option:
3
BUILD SUCCESSFUL (total time: 15 seconds)
```

Verifying Code (Array)

```
1 package simulationsep16;
2 import static java.lang.Character.toUpperCase;
3 import java.util.Arrays;
4 import javax.swing.JOptionPane;
5
6 public class SimulationSep16 {
7
8     public static void main(String[] args) {
9         char [] letters = {'A', 'B', 'C', 'D', 'E'};
10        String [] names = {"ody", "aya", "baleng", "fey", "leng"};
11
12        String inp = JOptionPane.showInputDialog(null, "Enter product code");
13        if (inp==null || inp.isEmpty()){
14            JOptionPane.showMessageDialog(null, "No Input Provided");
15            System.exit(0);
16        }
17        char Ucode = toUpperCase(inp.charAt(0));
18        int position = Arrays.binarySearch(letters, Ucode);
19
20        if (position >= 0) {
21            JOptionPane.showMessageDialog(null, "You entered " + inp + " which is " + names[position]);
22        } else {
23            JOptionPane.showMessageDialog(null, inp + " is invalid.");
24        }
25    }
26
27 }
```



Employee System (Inheritance)

```
1 package inheritanceactivity;
2 import javax.swing.JOptionPane;
3
4 class Employee {
5     String name;
6     int id;
7     double salary;
8
9     Employee(String name, int id, double salary) {
10         this.name = name;
11         this.id = id;
12         this.salary = salary;
13     }
14
15     void displayDetails() {
16         JOptionPane.showMessageDialog(null, "ID: " + id + "\nName: " + name + "\nSalary: " + salary);
17     }
18
19     void work() {
20         JOptionPane.showMessageDialog(null, name + " is workaholic.");
21     }
22 }
23
24 class Manager extends Employee {
25     int teamSize;
26
27     Manager(String name, int id, double salary, int teamSize) {
28         super(name, id, salary);
29         this.teamSize = teamSize;
30     }
31
32     @Override
33     void displayDetails() {
34         JOptionPane.showMessageDialog(null, "ID: " + id + "\nName: " + name + "\nSalary: " + salary + "\nTeam Size: " + teamSize);
35     }
36
37     @Override
38     void work() {
39         JOptionPane.showMessageDialog(null, name + " is manageholic.");
40     }
41 }
42
43 class Developer extends Employee {
44     String language;
45
46     Developer(String name, int id, double salary, String language) {
47         super(name, id, salary);
48         this.language = language;
49     }
50
51     @Override
52     void displayDetails() {
53         JOptionPane.showMessageDialog(null, "ID: " + id + "\nName: " + name + "\nSalary: " + salary + "\nProgramming Language: " + language);
54     }
55
56     @Override
57     void work() {
58         JOptionPane.showMessageDialog(null, name + " is codeholic in " + language);
59     }
60 }
61
62 public class InheritanceActivity {
63     public static void main(String[] args) {
64         int m = Integer.parseInt(JOptionPane.showInputDialog("How many employees do you wanna add?"));
65         Employee[] employees = new Employee[m];
66
67         for (int i = 0; i < m; i++) {
68             int job = Integer.parseInt(JOptionPane.showInputDialog("Enter type for employee " + (i + 1) + "\n1 > Employee\n2 > Manager\n3 > Developer"));
69             String name = JOptionPane.showInputDialog("Enter Name:");
70             int id = Integer.parseInt(JOptionPane.showInputDialog("Enter ID:"));
71             double salary = Double.parseDouble(JOptionPane.showInputDialog("Enter salary:"));
72
73             if (job == 2) {
74                 int teamSize = Integer.parseInt(JOptionPane.showInputDialog("Enter team size:"));
75                 employees[i] = new Manager(name, id, salary, teamSize);
76             } else if (job == 3) {
77                 String language = JOptionPane.showInputDialog("Enter programming language:");
78                 employees[i] = new Developer(name, id, salary, language);
79             } else {
80                 employees[i] = new Employee(name, id, salary);
81             }
82         }
83
84         for (Employee emp : employees) {
85             emp.displayDetails();
86             emp.work();
87         }
88     }
89 }
90
91
92 }
```


Input

How many employees do you wanna add?

1

OK Cancel

Input

Enter type for employee 1:

1 > Employee
2 > Manager
3 > Developer

3

OK Cancel

Input

Enter Name:

Melody

OK Cancel

Input

Enter ID:

101

OK Cancel

Input

Enter salary:

30000

OK Cancel

Input

Enter programming language:

Java

OK Cancel

Message

ID: 101
Name: Melody
Salary: 30000.0
Programming Language: Java

OK

Message

Melody is codeholic in Java

OK

ATM (Practical Exam)

